

BOR, Russian Federation

WMO: 238840

Lat: **61.5806N**Lon: **90.0331E**Elev: **58**StdP: **100.63**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.5	(c) -41.9	(d) -47.5	(e) 0.0	(f) -45.5	(g) -44.0	(h) 0.0	(i) -41.8	(j) 4.8	(k) -13.5	(l) 4.3	(m) -13.4	(n) 0.1	(o) 70	(p) 0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.6	(c) 29.5	(d) 19.1	(e) 27.4	(f) 18.4	(g) 25.1	(h) 17.6	(i) 20.6	(j) 26.6	(k) 19.5	(l) 25.0	(m) 18.4	(n) 23.4	(o) 2.2	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.6	22.7	17.6	12.7	21.6	16.5	11.8	20.7	59.9	26.9	56.2	25.0	52.6	23.5	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.9	4.3	3.8	-46.7	32.5	3.8	1.4	-49.5	33.6	-51.7	34.4	-53.8	35.2	-56.6	36.3
			-46.6	22.3	3.8	1.4	-49.3	23.3	-51.5	24.1	-53.7	24.9	-56.4	25.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.7	-24.2	-19.7	-9.6	-1.7	5.7	15.1	18.0	14.1	7.0	-1.7	-15.3	-21.6
	DBStd	16.14	10.57	8.62	7.39	5.89	5.32	5.02	3.66	3.90	4.39	5.83	9.90	10.55
	HDD10.0	5243	1061	832	609	351	156	14	1	8	111	364	758	979
	HDD18.3	7764	1319	1065	867	601	392	120	52	142	339	623	1008	1238
	CDD10.0	594	0	0	0	0	23	166	248	135	22	1	0	0
	CDD18.3	73	0	0	0	0	1	22	41	9	0	0	0	0
	CDH23.3	859	0	0	0	0	21	318	425	93	2	0	0	0
	CDH26.7	244	0	0	0	0	5	95	123	21	0	0	0	0
(14) Wind	WSAvg	1.5	1.2	1.3	1.7	2.0	1.9	1.6	1.5	1.4	1.6	1.7	1.5	1.3
(15) Precipitation	PrecAvg	589	38	27	27	32	48	57	65	73	65	60	53	44
	PrecMax	760	74	55	75	73	107	117	166	135	125	115	93	76
	PrecMin	389	2	6	2	8	18	21	7	13	11	15	20	16
	PrecStd	90	15	11	14	13	20	27	33	30	23	19	15	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.4	-0.5	8.2	15.9	25.8	31.8	32.4	29.2	22.2	12.4	2.5	-0.6
		MCWB	-1.9	-1.2	3.1	8.1	15.8	19.3	20.8	19.7	15.4	8.5	1.7	-1.1
	2%	DB	-4.7	-3.0	4.2	12.3	21.6	29.5	29.7	25.8	18.9	9.7	1.1	-2.7
		MCWB	-5.1	-3.8	0.6	6.1	13.3	18.6	19.7	18.5	13.6	7.4	0.8	-3.2
	5%	DB	-7.7	-5.6	1.8	9.3	18.4	27.2	27.8	23.3	16.0	7.6	0.0	-4.4
		MCWB	-8.2	-6.3	-0.5	4.6	11.5	17.7	18.9	17.2	12.2	5.7	-0.4	-4.8
	10%	DB	-10.3	-8.0	0.3	6.7	15.4	24.6	25.7	20.8	13.7	5.5	-1.6	-6.9
		MCWB	-10.8	-8.6	-1.5	3.1	9.8	16.7	18.3	16.1	11.2	4.1	-2.1	-7.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.7	-1.0	3.3	8.8	16.8	21.1	22.6	21.7	16.3	9.9	1.7	-0.9
		MCDB	-1.3	-0.6	6.9	14.4	25.0	28.1	29.1	27.0	20.6	11.7	2.1	-0.6
	2%	WB	-5.1	-3.8	1.1	6.8	14.2	19.7	21.1	19.4	14.4	7.6	0.8	-3.2
		MCDB	-4.7	-3.1	3.3	11.2	20.5	26.6	27.2	24.0	17.7	9.2	1.1	-2.8
	5%	WB	-8.2	-6.3	-0.1	5.0	12.2	18.6	20.1	17.9	12.9	6.0	-0.3	-4.8
		MCDB	-7.8	-5.6	1.7	8.9	17.4	24.8	25.5	22.0	15.5	7.4	0.0	-4.3
	10%	WB	-10.7	-8.6	-1.5	3.4	10.1	17.5	19.1	16.6	11.4	4.1	-2.0	-7.3
		MCDB	-10.3	-8.0	0.3	6.2	14.9	23.4	24.0	20.1	13.6	5.3	-1.6	-6.9
(35) Mean Daily Temperature Range	MDBR		7.7	9.6	11.2	10.3	10.3	11.4	10.6	8.9	7.1	5.3	6.6	7.3
	5% DB	MCDBR	7.1	9.0	11.0	13.5	15.9	16.5	14.9	13.1	11.5	7.4	5.4	7.3
		MCWBR	7.0	8.5	8.4	8.5	9.0	7.3	6.2	6.4	6.6	5.4	5.5	7.1
	5% WB	MCDBR	7.2	8.9	9.7	12.2	14.7	13.4	12.2	11.2	9.8	7.0	5.6	7.4
		MCWBR	7.1	8.6	7.7	8.1	8.5	6.7	5.9	6.1	6.1	5.4	5.6	7.2
(40) Clear-Sky Solar Irradiance	taub		0.266	0.304	0.340	0.365	0.382	0.391	0.451	0.411	0.357	0.320	0.258	0.240
	taud		1.914	1.971	1.975	2.040	2.243	2.371	2.173	2.272	2.390	2.235	1.970	1.870
	Ebn at Noon		378	619	741	811	835	835	763	765	748	620	388	231
	Edh at Noon		45	83	118	135	120	109	129	108	80	64	40	27
(44) All-Sky Solar Radiation	RadAvg		0.34	1.21	2.54	4.04	4.88	5.53	5.05	3.53	2.08	0.88	0.39	0.16
	RadStd		0.05	0.10	0.30	0.48	0.54	0.57	0.45	0.34	0.28	0.15	0.07	0.03

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.86	+3.09	+2.89	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page